RAJALAKSHMI ENGINEERING COLLEGE

Besh Prabanjeyan M 251001029-2025-IT D2

Started on	Tuesday, 15 September 2026, 9:36 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 9:37 AM
Time taken	1 min 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int max(int a, int b)
{
    return (a > b) ? a : b;
}
int main()
{
    int n;
    scanf("%d", &n);
    int a[n];
    int dp[n];
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
        dp[i] = 1;
    }
    for (int i = 1; i < n; i++)
    {
        for (int j = 0; j < i; j++)
        {
            if (a[j] <= a[i])
            {
                dp[i] = max(dp[i], dp[j] + 1);
            }
        }
    }
    int answer = 0;
    for (int i = 0; i < n; i++)
    {
        if (dp[i] > answer)
        {
            answer = dp[i];
        }
    }
    printf("%d\n", answer);
    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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